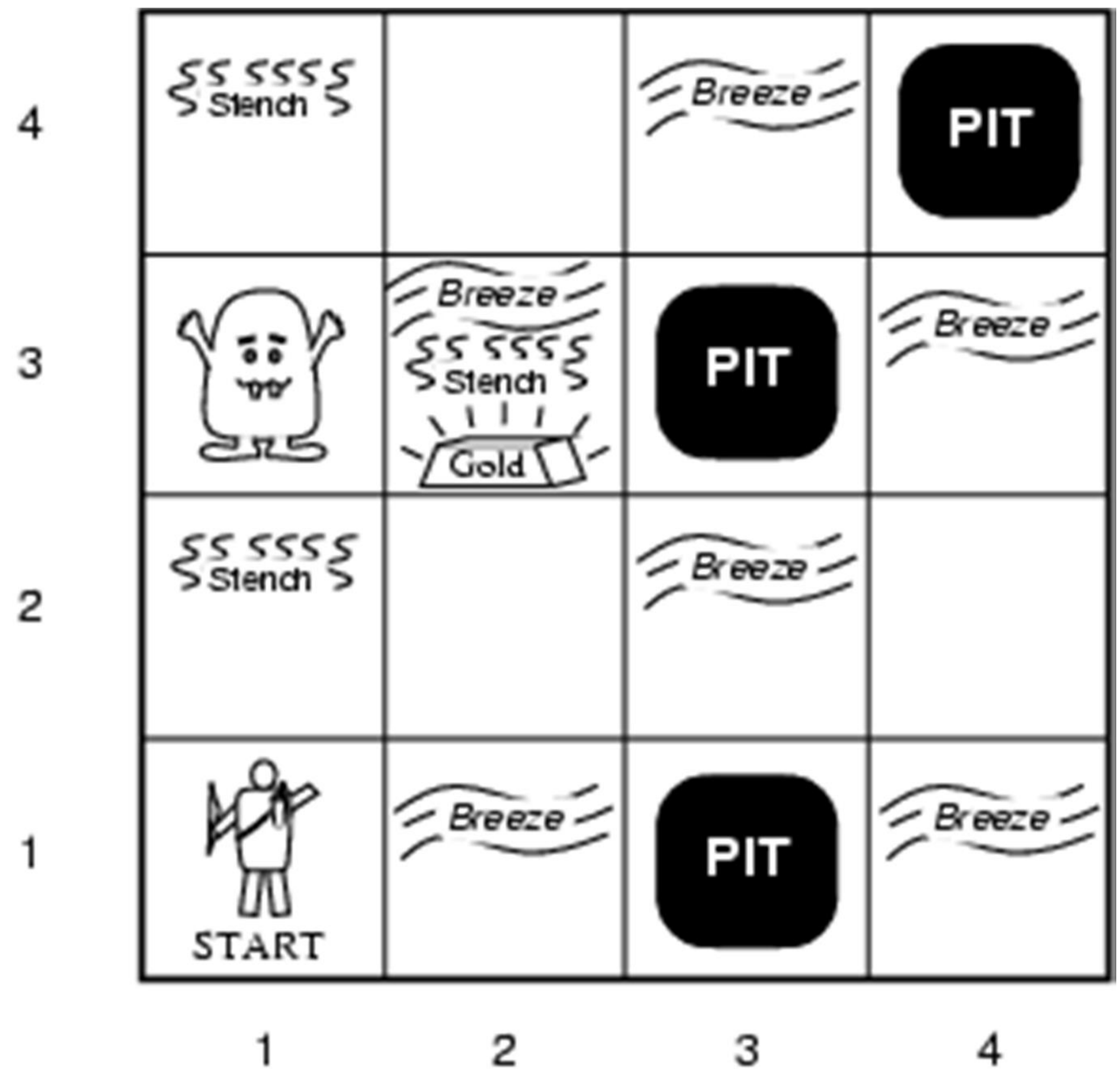
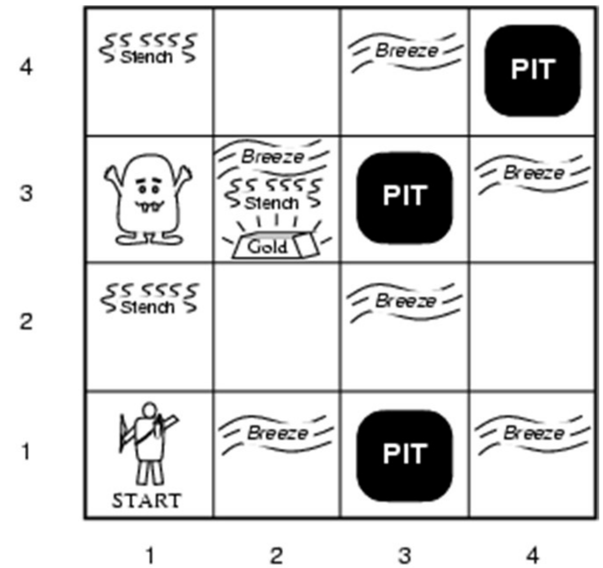
A Wumpus World: One more Example but complex scenario



Wumpus World PEAS description

- Performance measure
  - gold +1000, death -1000
  - -1 per step, -10 for using the arrow

- Environment: 4 x 4 grid of rooms
  - Squares adjacent to wumpus are smelly
  - Squares adjacent to pit are breezy
  - Glitter iff gold is in the same square
  - Shooting kills wumpus if you are facing it
  - Shooting uses up the only arrow
  - Grabbing picks up gold if in same square
  - Releasing drops the gold in same square



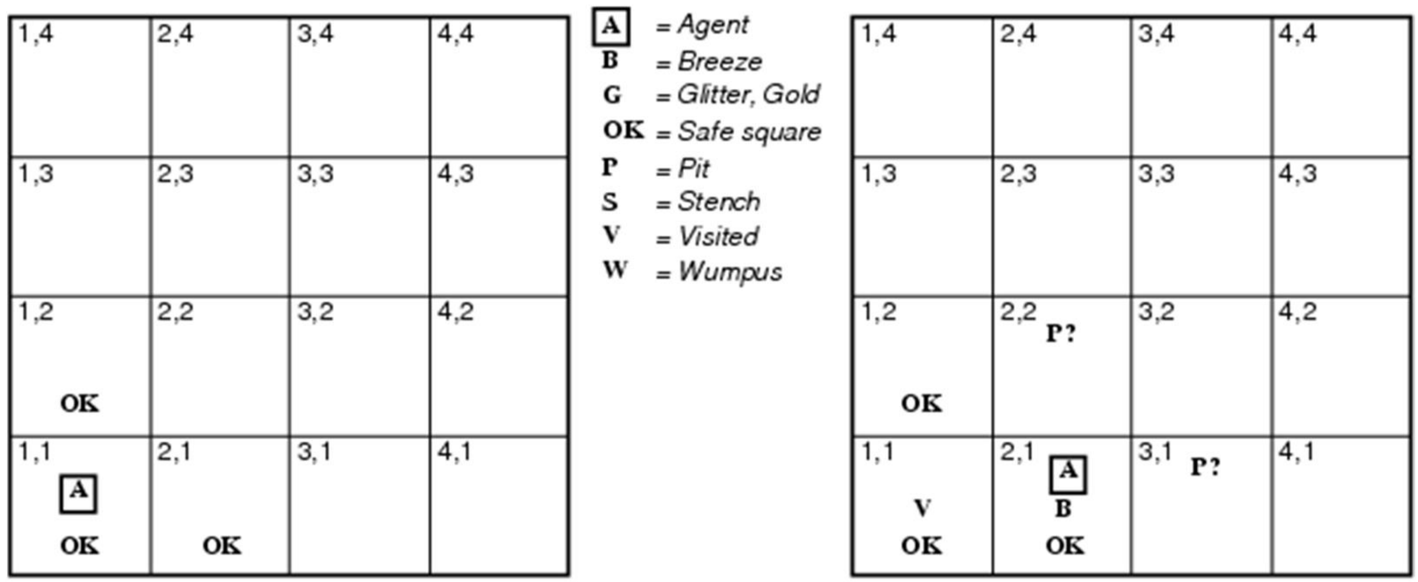
- Sensors: Stench, Breeze, Glitter, Bump, Scream (shot Wumpus)
- Actuators: Left turn, Right turn, Forward, Grab, Release, Shoot

Reasoning in the Wumpus World

- Agent has initial ignorance about the configuration
  - Agent knows his/her initial location
  - Agent knows the rules of the environment

- Goal is to explore environment, make inferences (reasoning) to try to find the gold.

Exploring the Wumpus World



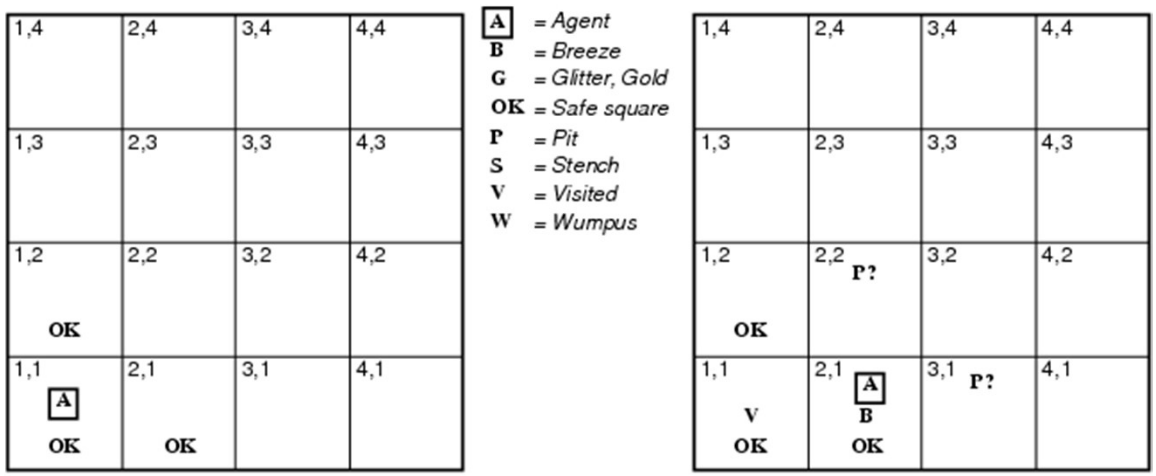
(a) (b)

[1,1] The KB initially contains the rules of the environment.

The first percept is [*none, none,none,none,none*],

move to safe cell e.g. 2,1

Exploring the Wumpus World



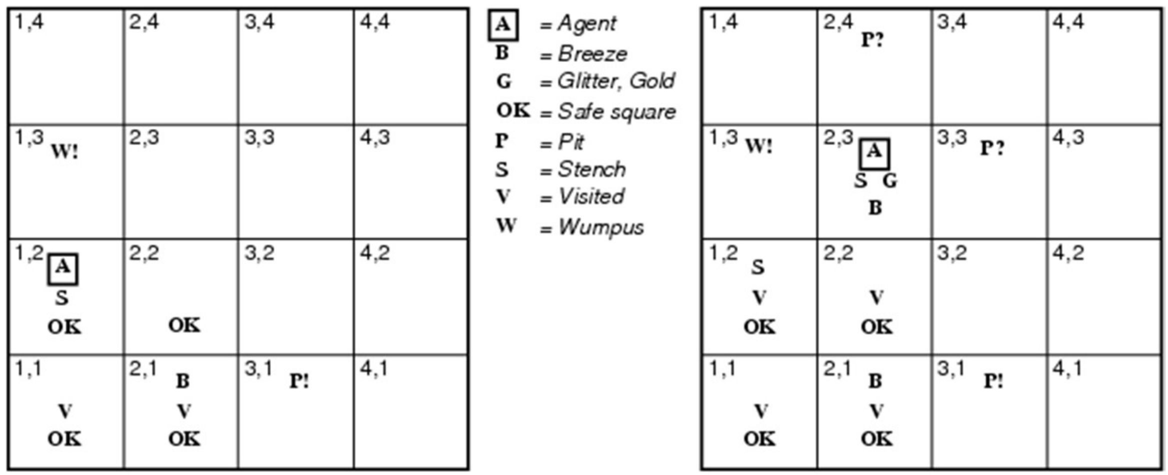
(a) (b)

[2,1] = breeze

indicates that there is a pit in [2,2] or [3,1],

return to [1,1] to try next safe cell

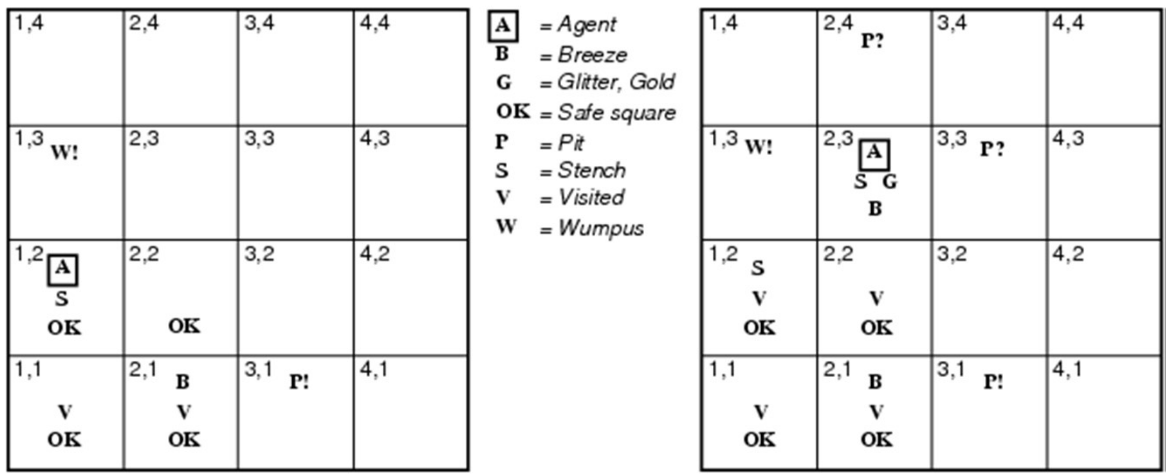
Exploring the Wumpus World



(a) (b)

[1,2] Stench in cell which means that wumpus is in [1,3] or [2,2]
YET … not in [1,1]
YET … not in [2,2] or stench would have been detected in [2,1]
(this is relatively sophisticated reasoning!)

Exploring the Wumpus World



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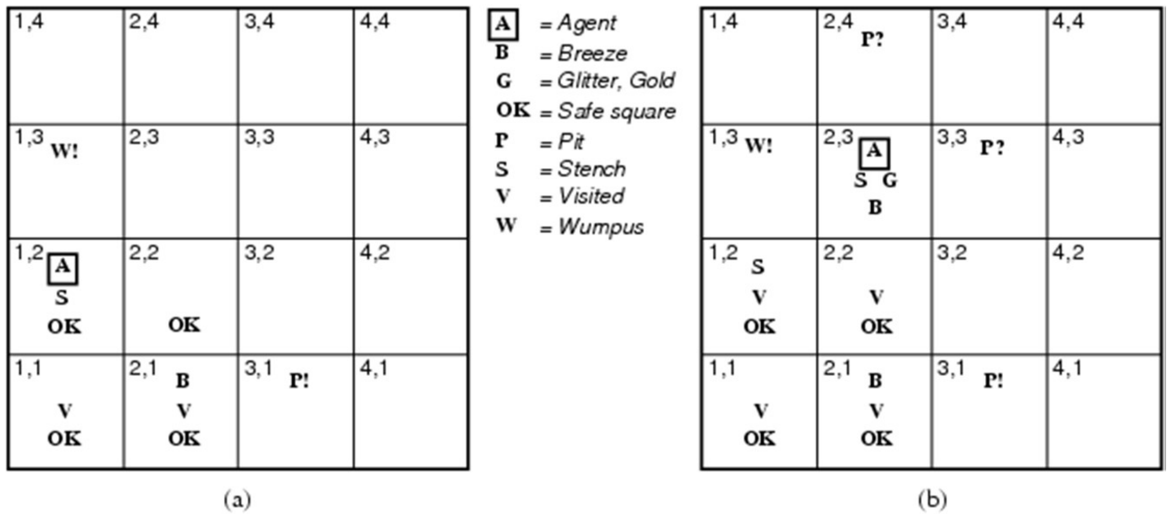
THUS … wumpus is in [1,3]
THUS [2,2] is safe because of lack of breeze in [1,2]
THUS pit in [1,3] (again a clever inference)
move to next safe cell [2,2]

Exploring the Wumpus World



[2,2] move to [2,3]

[2,3] detect glitter , smell, breeze
THUS pick up gold
THUS pit in [3,3] or [2,4]

A simple knowledge base: Wumpus world

- $P_{x,y}$ is true if there is a pit in [x, y].
- $W_{x,y}$ is true if there is a wumpus in [x, y], dead or alive.
- $B_{x,y}$ is true if the agent perceives a breeze in [x, y].
- $S_{x,y}$ is true if the agent perceives a stench in [x, y].

Wumpus world sentences

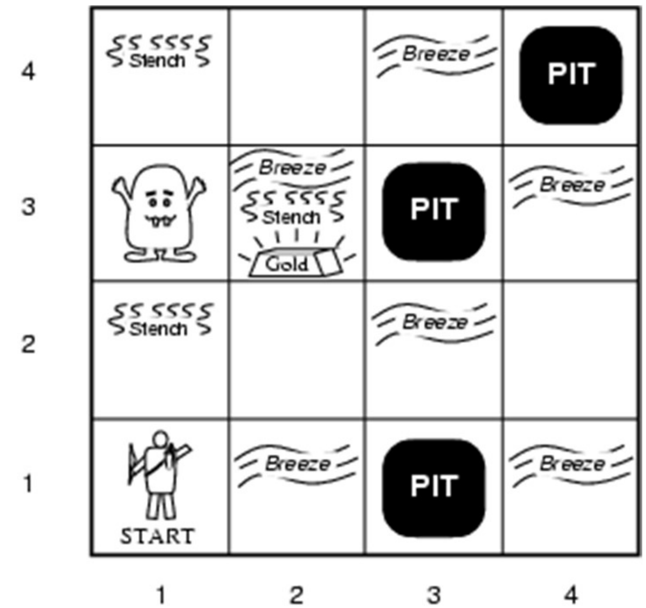
Let $P_{i,j}$ be true if there is a pit in [i, j].
Let $B_{i,j}$ be true if there is a breeze in [i, j].

start:

R1: $\neg P_{1,1}$

R4: $\neg B_{1,1}$

R5: $B_{2,1}$



- "Pits cause breezes in adjacent squares"

  R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

  R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

- KB can be expressed as the conjunction of all of these sentences

Inference in Wumpus World

- Let us see how these inference rules and equivalences can be used in the wumpus world.
- We start with the knowledge base containing R1 through R5 and show how to prove $\neg P_{1,2}$, that is, there is no pit in [1,2].

First, we apply biconditional elimination to R2 to obtain

- R6: $(B1,1 \Rightarrow (P1,2 \lor P2,1)) \land ((P1,2 \lor P2,1) \Rightarrow B1,1)$.

Then we apply And-Elimination to R6 to obtain

- R7: $((P1,2 \lor P2,1) \Rightarrow B1,1)$.

Logical equivalence for contrapositives gives

- R8: $(\neg B1,1 \Rightarrow \neg(P1,2 \lor P2,1))$.

Now we can apply Modus Ponens with R8 and the percept R4 (i.e., $\neg B1,1$), to obtain

- R9 : $\neg(P1,2 \lor P2,1)$.

Finally, we apply De Morgan's rule, giving the conclusion

- R10 : $\neg P1,2 \land \neg P2,1$.

That is, neither [1,2] nor [2,1] contains a pit.

Exercise

Natural Language Context:

1. If a student is a senior, then they are eligible for advanced seminars.
2. If a student is eligible for advanced seminars, then they have completed all core courses.
3. If a student has completed all core courses, then they have taken either Advanced Mathematics or Advanced Physics.
4. If a student has taken Advanced Mathematics, then Professor Smith was their instructor.

5. If Professor Smith was a student's instructor, then the student has participated in the annual Math Olympiad.

6. If a student has taken Advanced Physics, then they have access to the quantum computing lab.

7. If a student has access to the quantum computing lab, then they are working on a research project.

8. If a student is working on a research project, then they have a faculty advisor.

9. If a student has a faculty advisor, then they have maintained a GPA of at least 3.5.

10. Alice is a senior student.

11. Alice has not taken Advanced Mathematics.

Conclusion: Alice has maintained a GPA of at least 3.5